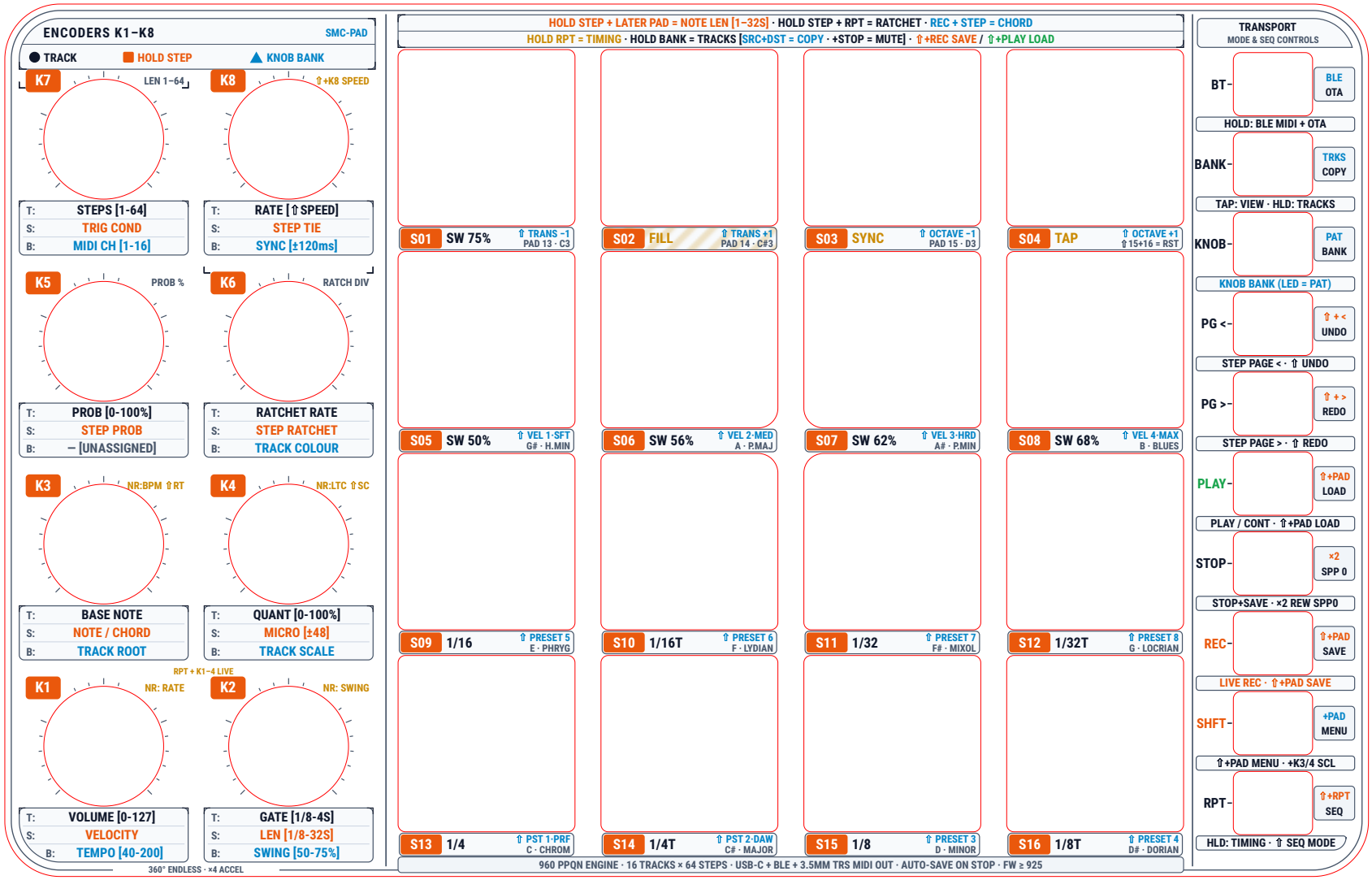

ENCODERS K1–K8
SMC-PAD
TRACK
HOLD STEP
KNOB BANK
K7
LEN 1–64
T: STEPS [1-64]
S: TRIG COND
B: MIDI CH [1-16]
K8
⇧+K8 SPEED
T: RATE [⇧SPEED]
S: STEP TIE
B: SYNC [±120ms]
K5
PROB %
T: PROB [0-100%]
S: STEP PROB
B: — [UNASSIGNED]
K6
RATCH DIV
T: RATCHET RATE
S: STEP RATCHET
B: TRACK COLOUR
K3
NR:BPM ⇧RT
T: BASE NOTE
S: NOTE / CHORD
B: TRACK ROOT
K4
NR:LTC ⇧SC
T: QUANT [0-100%]
S: MICRO [±48]
B: TRACK SCALE
RPT + K1–4 LIVE
K1
NR: RATE
T: VOLUME [0-127]
S: VELOCITY
B: TEMPO [40-200]
K2
NR: SWING
T: GATE [1/8-4S]
S: LEN [1/8-32S]
B: SWING [50-75%]
360° ENDLESS · ×4 ACCEL
HOLD STEP + LATER PAD = NOTE LEN [1–32S] · HOLD STEP + RPT = RATCHET · REC + STEP = CHORD
HOLD RPT = TIMING · HOLD BANK = TRACKS [SRC+DST = COPY · +STOP = MUTE] · ⇧+REC SAVE / ⇧+PLAY LOAD
S01 SW 75% ⇧ TRANS −1 PAD 13 · C3
S02 FILL ⇧ TRANS +1 PAD 14 · C#3
S03 SYNC ⇧ OCTAVE −1 PAD 15 · D3
S04 TAP ⇧ OCTAVE +1 ⇧15+16 = RST
S05 SW 50% ⇧ VEL 1·SFT G# · H.MIN
S06 SW 56% ⇧ VEL 2·MED A · P.MAJ
S07 SW 62% ⇧ VEL 3·HRD A# · P.MIN
S08 SW 68% ⇧ VEL 4·MAX B · BLUES
S09 1/16 ⇧ PRESET 5 E · PHRYG
S10 1/16T ⇧ PRESET 6 F · LYDIAN
S11 1/32 ⇧ PRESET 7 F# · MIXOL
S12 1/32T ⇧ PRESET 8 G · LOCRIAN
S13 1/4 ⇧ PST 1·PRF C · CHROM
S14 1/4T ⇧ PST 2·DAW C# · MAJOR
S15 1/8 ⇧ PRESET 3 D · MINOR
S16 1/8T ⇧ PRESET 4 D# · DORIAN
960 PPQN ENGINE · 16 TRACKS × 64 STEPS · USB-C + BLE + 3.5MM TRS MIDI OUT · AUTO-SAVE ON STOP · FW ≥ 925
TRANSPORT
MODE & SEQ CONTROLS
BT
BLE OTA
HOLD: BLE MIDI + OTA
BANK
TRKS COPY
TAP: VIEW · HLD: TRACKS
KNOB
PAT BANK
KNOB BANK (LED = PAT)
PG <
⇧ + < UNDO
STEP PAGE < · ⇧ UNDO
PG >
⇧ + > REDO
STEP PAGE > · ⇧ REDO
PLAY
⇧+PAD LOAD
PLAY / CONT · ⇧+PAD LOAD
STOP
×2 SPP 0
STOP+SAVE · ×2 REW SPP0
REC
⇧+PAD SAVE
LIVE REC · ⇧+PAD SAVE
SHFT
+PAD MENU
⇧+PAD MENU · +K3/4 SCL
RPT
⇧+RPT SEQ
HLD: TIMING · ⇧ SEQ MODE

SMC-PAD // SHEET 2 OF 2: 360° PINWHEEL SIDE BANDS & I/O WELL INSERT — ALIGN FRAME: 250.00 × 165.00 MM

CNR-BR

BTN SIDE

SEAM ⟶ ▶

R10.7 · 90°

REAR-01 // HARDWARE SCHEMATIC [JIELI AC636N · BR25]

- 16× VEL PADS — MUX PB2/3 → PA4/6
- 8× ENC + 10× BTN — SPI2 595 → PB0/PC1
- 3.7V LI-ION BAT — 704060 · 2000mAh

AC636N · BR25
pi32v2 · 960 PPQN
PI32V2 · 512K NOR
RAM ISR SAFE

- 16× WS2812 RGB — SPI1 PB1 · 800kHz
- TRS MIDI OUT — UART1 PC5 · 31.25k
- USB-C + BLE MIDI — 3-CBL · AE40 OTA

REAR-02 // FLASH MAP, SYSEX & SAFE-MODE RECOVERY

FLASH 512K:	0x00000 CODE · 0x30000 16×PAT · 0x50000 VM · 0x77000 PST
USB 3-CBL:	CBL0: Master MIDI · CBL1: Private Sinco · CBL2: Mackie
DIAG SYSEX:	F0 7D 53 50 01/02/05 F7 (status / counters / crash dump)
SAFE MODE:	Power-cycle 2× in 5s → skips pads/keys, keeps BLE + OTA!
UBOOT ROM:	USB_KEY handshake on DP/DM enters BR25 Mask-ROM loader

PWR · CHG · USB-C · MIDI OUT

5V ⎓ 500mA · RED=CHG / GRN=FULL · TRS-A MIDI OUT

▲ REAR PANEL (222.43×23.80mm) ▼ FRONT CHEAT SHEET (222.43×23.80mm)

CNR-FL

KNOB SIDE

⇧+RPT=SEQ

FW ≥ 925

F1 // STEP & NOTE LENGTH

TAP PAD:	Step On / Off (S01..S16)
HOLD+PAD:	Note Length 1–32 steps!
HOLD+KNOB:	Step P-Lock (K3 moves chord)
HOLD+RPT:	Ratchet ×1 → ×2 → ×3 → ×4
PG < / >:	Page 1–4 · ⇧+< / > Undo/Redo

F2 // TRACKS [HOLD BANK]

TAP BANK:	Step View ↔ Play View
HOLD+PAD:	Select active Track 1–16
SRC+DST:	Hold Src+Tap Dst = Copy Notes
+⇧ COPY:	Full Copy (+Ch / Note / Vol)
+STOP+PAD:	Mute Trk · +⇧+STOP = Clear

F3 // LIVE REC & TIMING

REC+PLAY:	Live Rec (hold = gate len)
CHORDS:	Hit active step (up to 4 notes)
QUANTIZE:	K4 (0–100%) non-destructive!
HOLD RPT:	P1-8 Rate · P9-13 Swg · P14 Fill
SYNC/TAP:	P15 Ext Clock · P16 Tap Tempo

F4 // 16 PATTERNS & CUE

⇧+REC+PAD:	Save Slot 1–16 (no pad=cur)
⇧+PLAY+PAD:	Load Slot (cues next bar!)
SLOT LEDS:	Off=Empty · Dim=Used · Lit=Cur
FEEDBACK:	Blinks 3× GRN=OK / RED=ERR
AUTO-SAVE:	STOP saves edits to flash

▼ LEFT ENCODER SIDE (142.43×23.80mm)

CNR-BL

REAR SEAM

SEAM ⟶ ▶

R10.7 · 90°

LEFT-01 // K1–K8 KNOB MATRIX

K1	Vol 0–127	Velocity	BPM 40–200
K2	Gate 1/8–4	Len 1/8–32	Swing 50–75
K3	Base Note	Note/Chord	Trk Root
K4	Quantize	Micro ±48	Trk Scale
K5	Prob %	Step Prob	—
K6	Ratch Rate	Stp Ratch	Trk Colour
K7	Steps 1–64	Trig Cond	MIDI Ch
K8	Grid Rate	Step Tie	Sync ±120ms

LEFT-02 // TRIG, GATE & RATCHET

TRIG 1–6:	P1:ALW · P2-3: 1:2, 2:2 · P4-6: 1..3:3
TRIG 7–13:	P7-10: 1..4:4 · P11:1ST · P12:FILL · 13:!F
GATE [K2]:	1/8..7/8, 1..4S · Step Len: 1..32S
RATCH [K6]:	AUTO(split) · 1/8 · 1/16 · 16T · 32 · 64
GRID [K8]:	Keeps timing (dim=off-grid) · ⇧+K8 Speed

▼ REAR I/O WELL INSERTS (PRIMARY + SPARE, 46.00×12.80mm)

POWER · CHG · USB-C · MIDI

POWER · CHG · USB-C · MIDI

▼ RIGHT TRANSPORT SIDE (142.43×23.80mm)

CNR-FR

FRONT SEAM

STOP×2=0

R10.7 · 90°

RIGHT-01 // ROOTS & SCALES [P1–12]

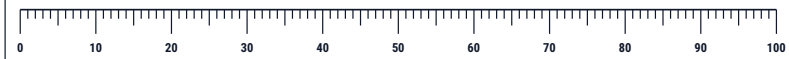
P01: C · Chromatic	P07: F# · Mixolydian
P02: C# · Major	P08: G · Locrian
P03: D · Nat Minor	P09: G# · Harm Minor
P04: D# · Dorian	P10: A · Pent Major
P05: E · Phrygian	P11: A# · Pent Minor
P06: F · Lydian	P12: B · Blues

0 LIT = FOLLOW · BANK K3/4 = TRK · ⇧+K3/4 = GLOBAL

RIGHT-02 // LED READOUTS & CLOCK

BAR LEDS:	Vol / Vel / Prob / Qnt / BPM: P1→16
SIGNED:	Micro & Sync: P8..1 (-) · P9..16 (+)
NOTE KEY:	P1–12 = C..B Root · P13–16 = Octave
EXT SYNC:	P15 On = 24 PPQN · STOP×2 = SPP 0
SYNC MS:	BANK K8: −120..+120ms (Master/Slave)

SCALE CHECK: 100.00 MM VERNIER



360° PINWHEEL WRAP:

1. Top Deck (flush inside rim)
2. Rear → Left (butt at SEAM)
3. Front → Right (butt at SEAM)
4. I/O Insert (inside rear well)

1:1 SCALE · US LETTER 8.5×11

CAMEO 5: Cut RED (#ff0000) paths · Set BLUE (#0066ff) 250×165mm Align Frame to NO CUT after centering.